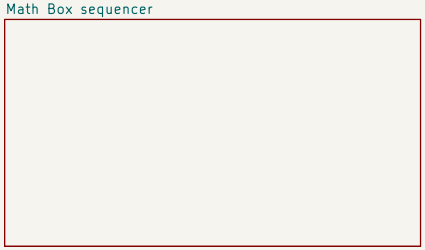
Address decoder



File: dec.kicad_sch

Math Box sequencer



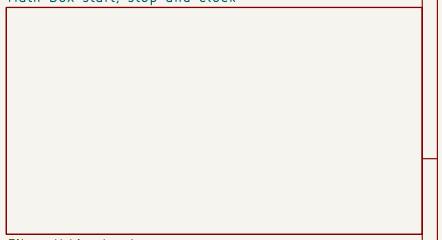
File: seq.kicad_sch

Math Box ALU



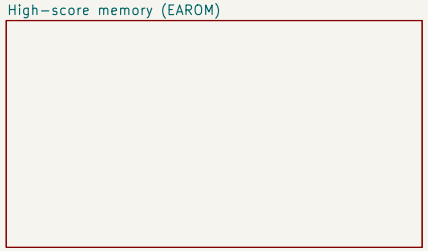
File: alu.kicad_sch

Math Box start, stop and clock



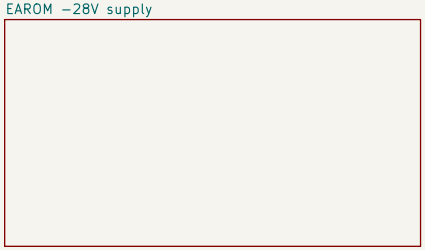
File: ctl.kicad_sch

High-score memory (EAROM)



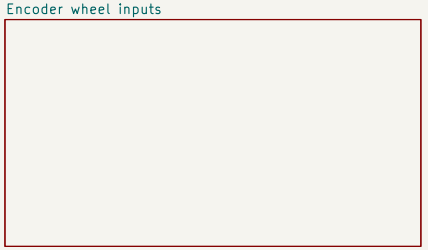
File: ear.kicad_sch

EAROM -28V supply



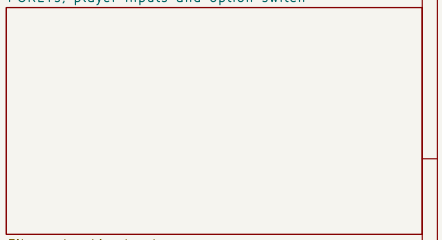
File: eps.kicad_sch

Encoder wheel inputs



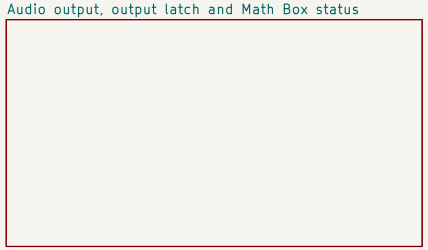
File: spin.kicad_sch

POKEYs, player inputs and option switch



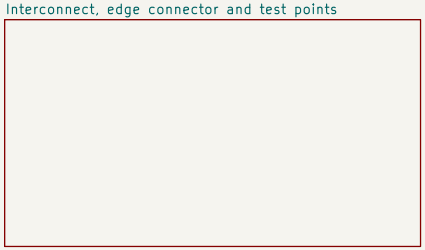
File: pokey.kicad_sch

Audio output, output latch and Math Box status



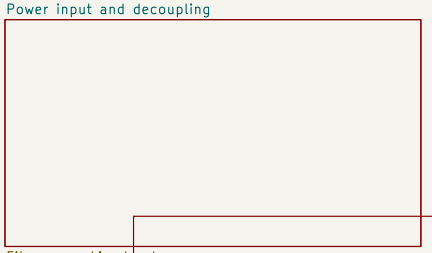
File: audio.kicad_sch

Interconnect, edge connector and test points



File: io.kicad_sch

Power input and decoupling

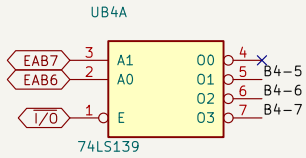


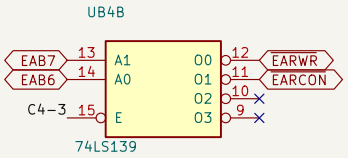
File: power.kicad_sch

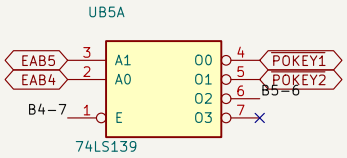
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File: tempest-auxiliary.kicad_sch

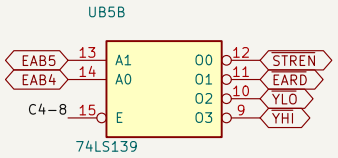
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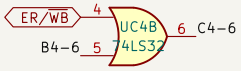


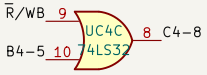


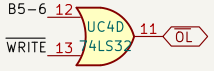


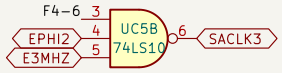


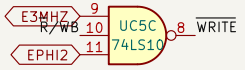


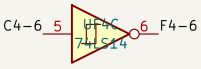


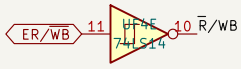


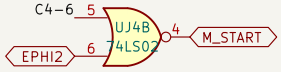










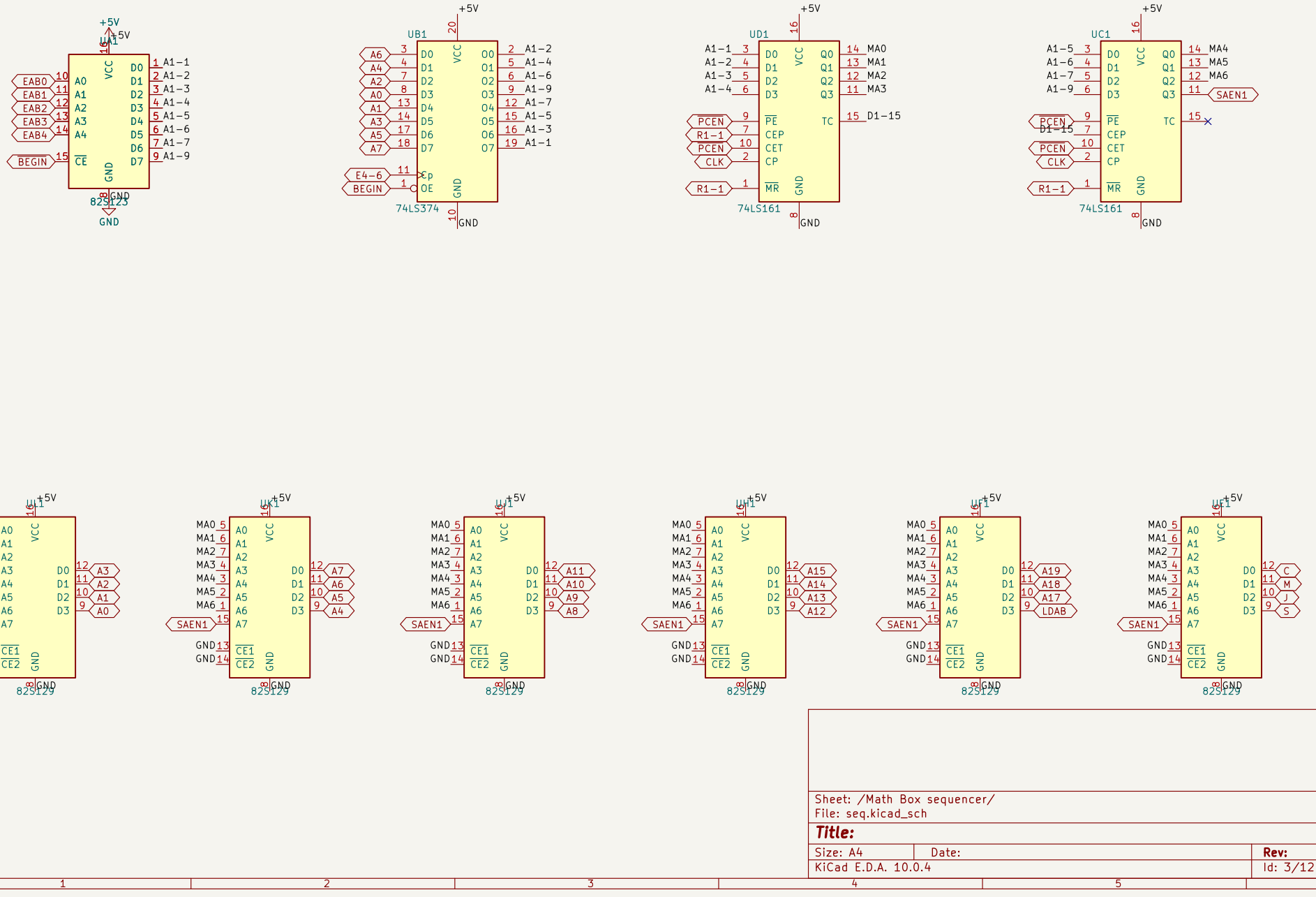


Sheet: /Address decoder/
File: dec.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

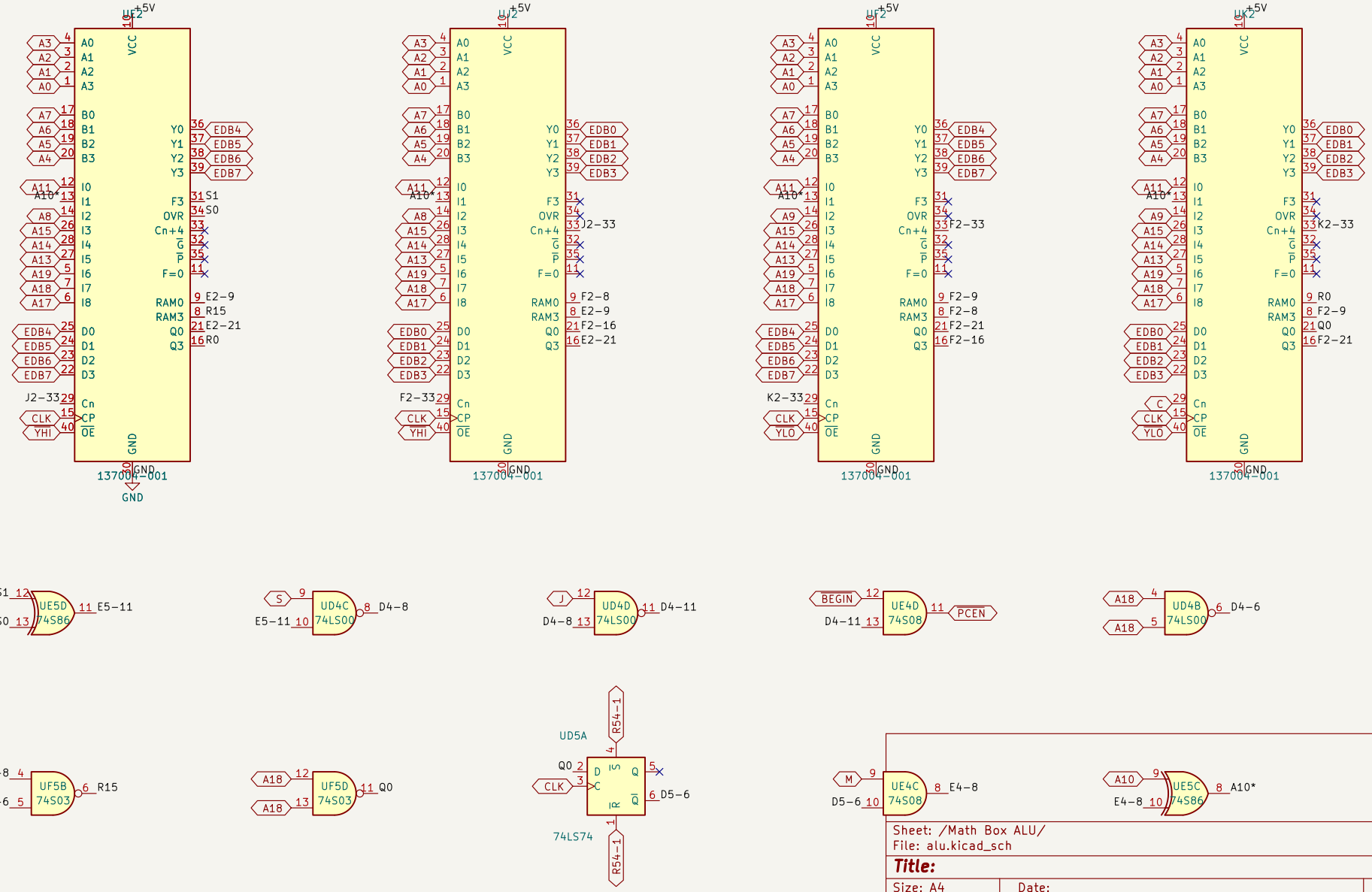
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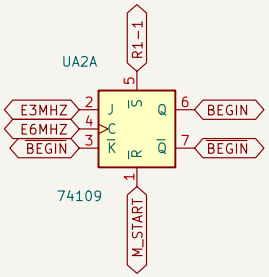


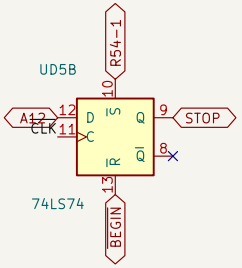
Sheet: /Math Box sequencer/
File: seq.kicad_sch

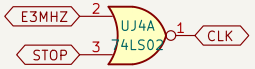
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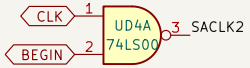

UF2
+5V
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10* A11 A12 A13 A14 A15
B0 B1 B2 B3 I0 I1 I2 I3 I4 I5 I6 I7 I8
D0 D1 D2 D3 Cn CP OE
Y0 Y1 Y2 Y3 F3 OVR Cn+4 G P F=0 RAM0 RAM3 Q0 Q3
VCC GND
CLK YHI YLO C
EDB0 EDB1 EDB2 EDB3 EDB4 EDB5 EDB6 EDB7
S1 S0 J2-33 F2-33 K2-33 E2-9 R15 E2-21 R0 F2-8 E2-16 F2-9 F2-21 F2-16 Q0
137004-001
UE5D 74S86
UD4C 74LS00
UD4D 74LS00
UE4D 74S08
UD4B 74LS00
UF5B 74S03
UF5D 74S03
UD5A
74LS74
R54-1
UE4C 74S08
UE5C 74S86
R15
R42
+5V
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Title:
Size: A4
Date:
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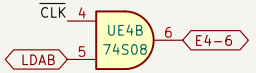


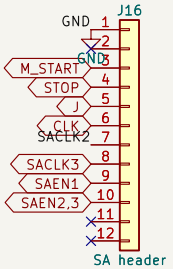








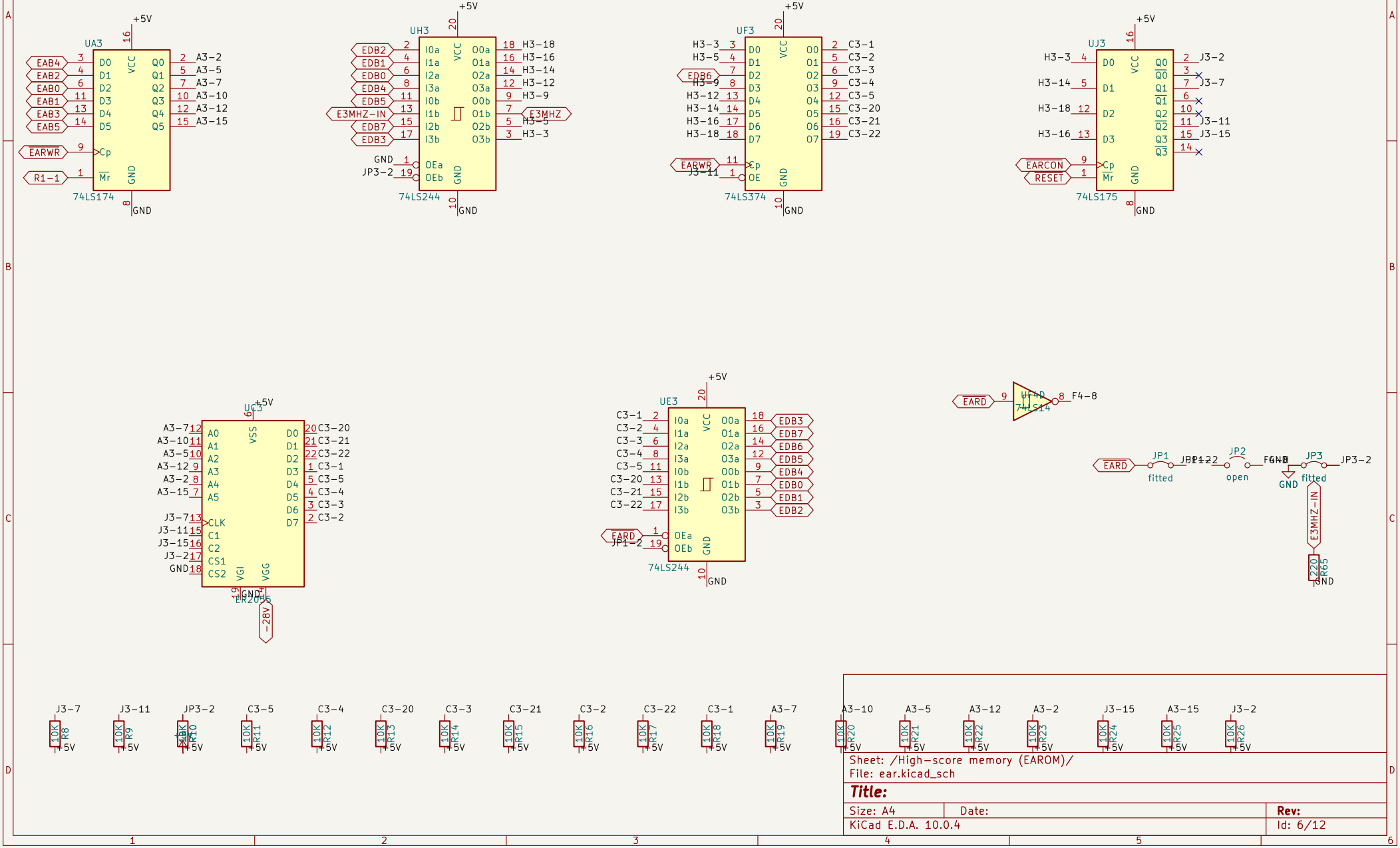


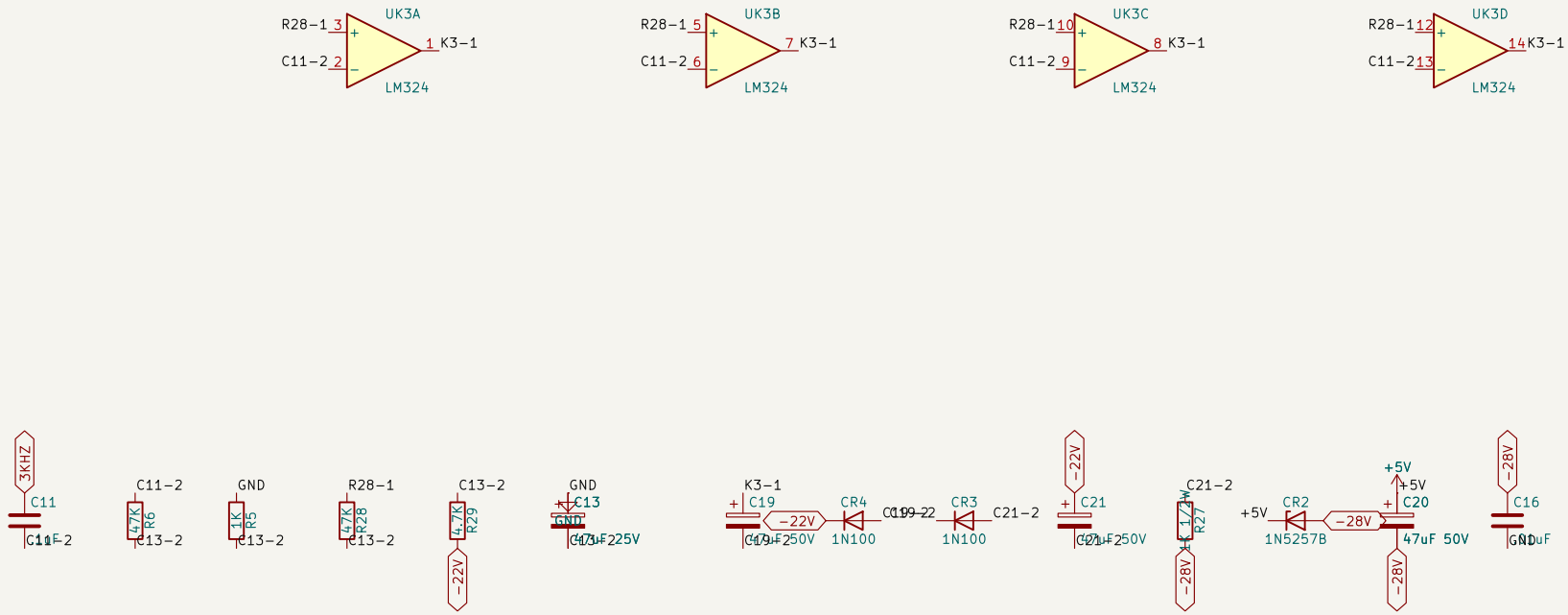


Sheet: /Math Box start, stop and clock/
File: ctl.kicad_sch

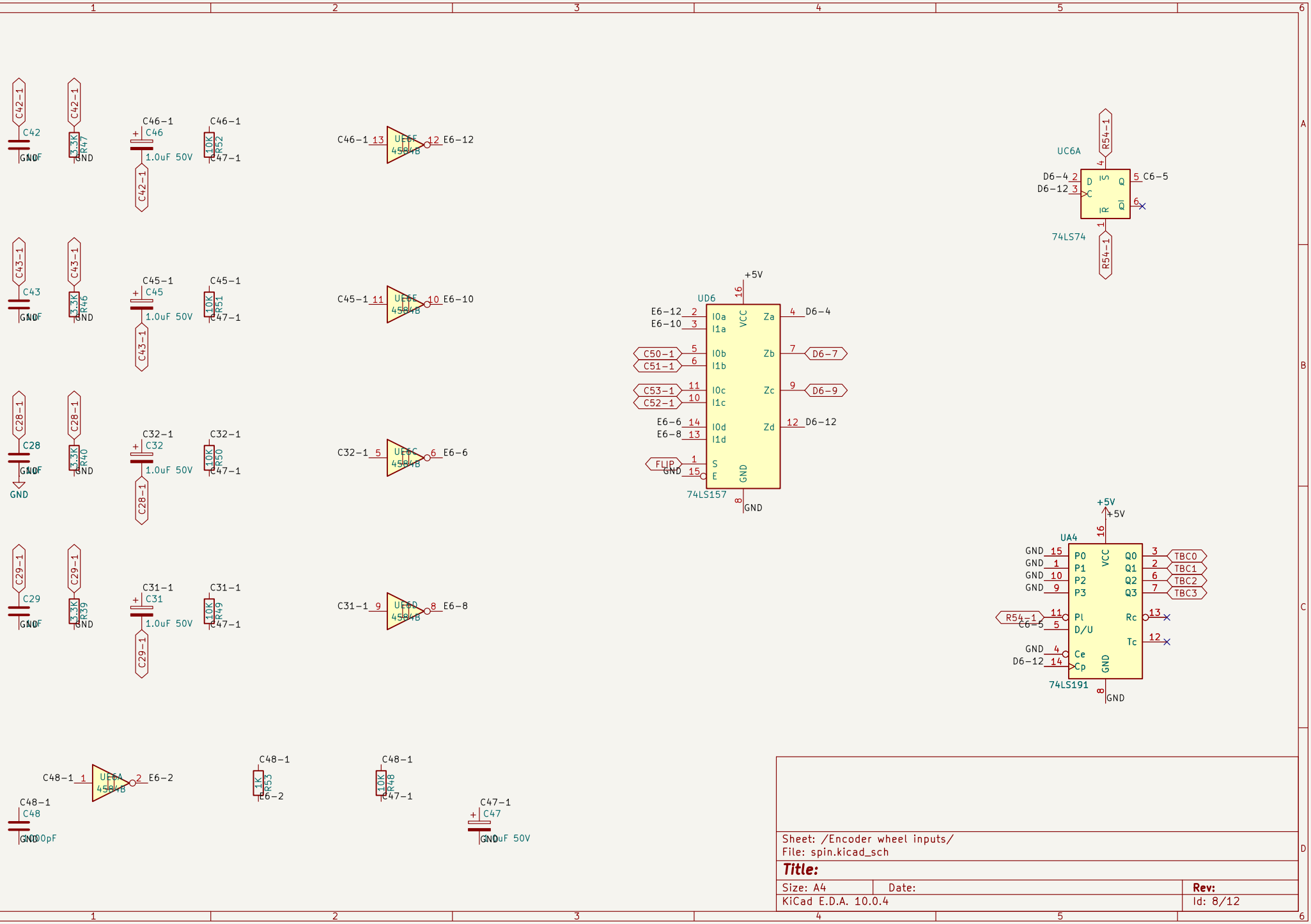
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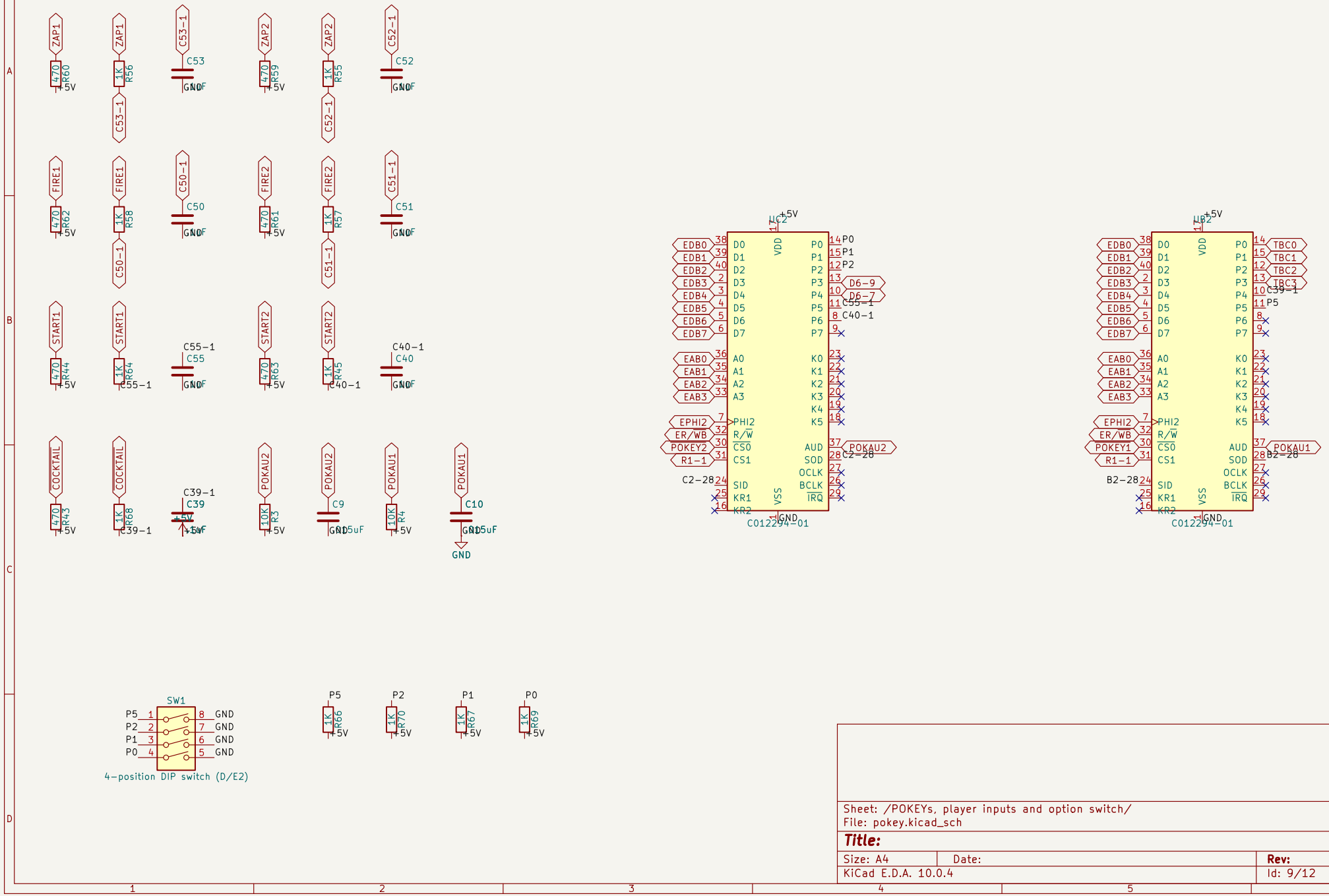
Size: A4	Date:	**Rev:**
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UA3
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UH3
74LS244
UF3
74LS374
UJ3
74LS175
UC3
ER2055
UE3
74LS244
UF4D
74LS14
JP1
JP2
JP3
fitted
open
fitted
E3MHZ-IN
R65
220
Sheet: /High-score memory (EAROM)/
File: ear.kicad_sch
Title:
Size: A4
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Sheet: /EAROM −28V supply/		
File: eps.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
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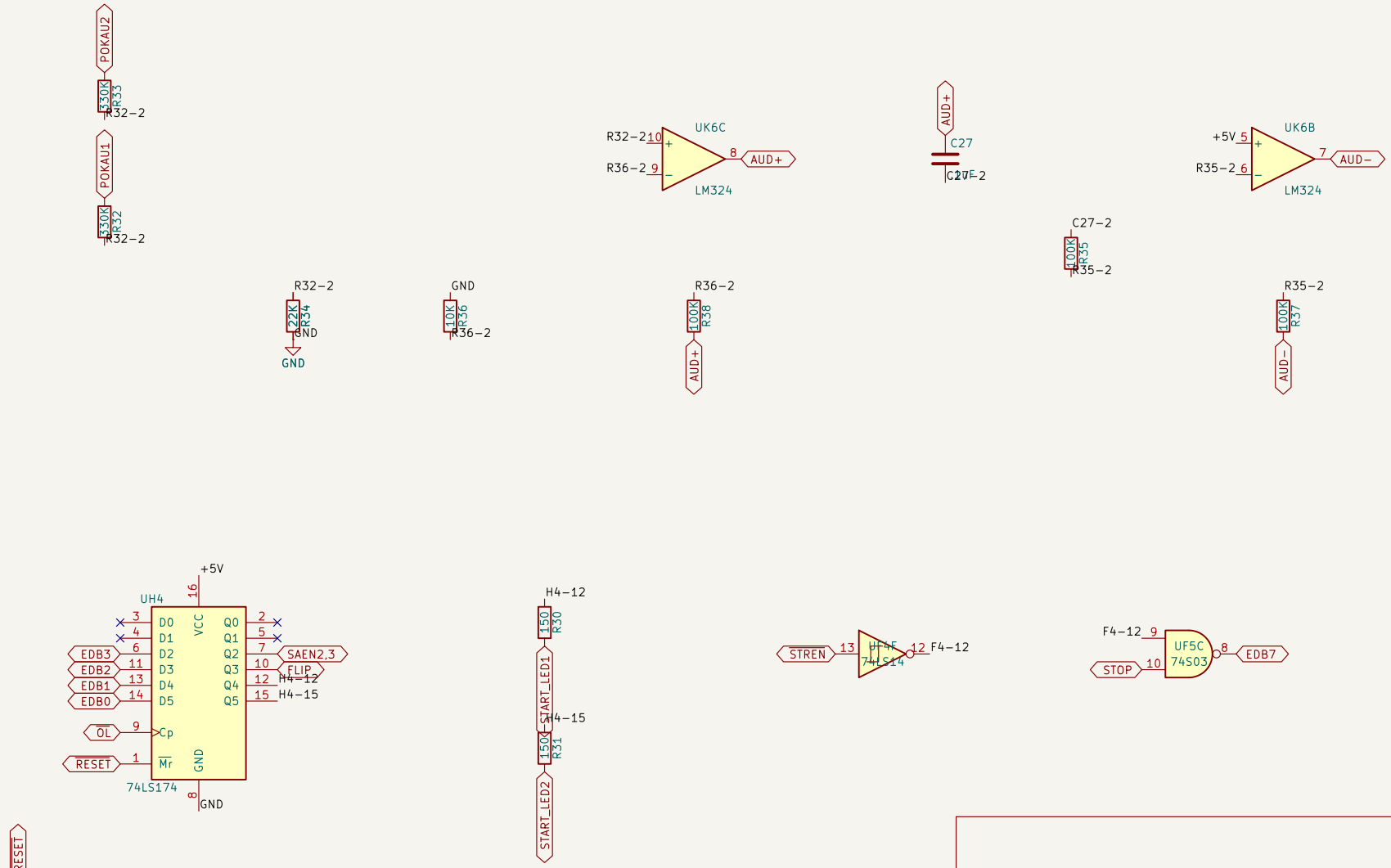

C42-1
C42
100pF
GND
C42-1
3.3K
R47
GND
C46-1
C46
1.0uF 50V
C42-1
C46-1
10K
R52
C47-1
C46-1 13 U6F 12 E6-12
C43-1
C43
100pF
GND
C43-1
3.3K
R46
GND
C45-1
C45
1.0uF 50V
C43-1
C45-1
10K
R51
C47-1
C45-1 11 U6E 10 E6-10
C28-1
C28
100pF
GND
C28-1
3.3K
R40
GND
C32-1
C32
1.0uF 50V
C28-1
C32-1
10K
R50
C47-1
C32-1 5 U6C 6 E6-6
C29-1
C29
100pF
GND
C29-1
3.3K
R39
GND
C31-1
C31
1.0uF 50V
C29-1
C31-1
10K
R49
C47-1
C31-1 9 U6D 8 E6-8
C48-1 1 U6A 2 E6-2
C48-1
C48
100pF
GND
C48-1
1K
R53
E6-2
C48-1
10K
R48
C47-1
C47-1
C47
1.0uF 50V
GND
+5V
UD6
16
E6-12 2 I0a
E6-10 3 I1a
C50-1 5 I0b
C51-1 6 I1b
C53-1 11 I0c
C52-1 10 I1c
E6-6 14 I0d
E6-8 13 I1d
FLIP 1 S
GND 15 E
VCC
GND
8
Za 4 D6-4
Zb 7 D6-7
Zc 9 D6-9
Zd 12 D6-12
74LS157
GND
UC6A
R54-1
4
D6-4 2 D
D6-12 3 C
S
R
Q 5 C6-5
Q 6
1
74LS74
R54-1
+5V
+5V
UA4
16
GND 15 P0
GND 1 P1
GND 10 P2
GND 9 P3
R54-1 11 Pl
C6-5 5 D/U
GND 4 Ce
D6-12 14 Cp
VCC
GND
8
Q0 3 TBC0
Q1 2 TBC1
Q2 6 TBC2
Q3 7 TBC3
Rc 13
Tc 12
74LS191
GND
Sheet: /Encoder wheel inputs/
File: spin.kicad_sch
Title:
Size: A4
Date:
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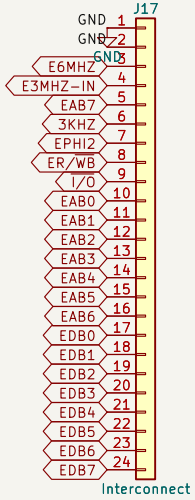


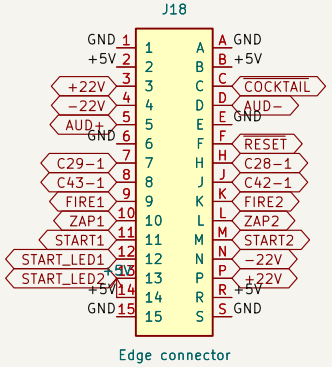
Sheet: /POKEYs, player inputs and option switch/
File: pokey.kicad_sch

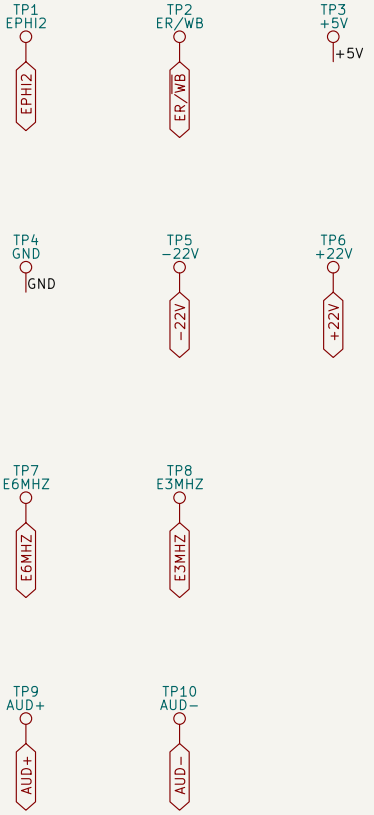
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POKAU2
330K
R33
R32-2
POKAU1
330K
R32
R32-2
R32-2
22K
R34
GND
GND
GND
10K
R36
R36-2
R36-2
100K
R38
AUD+
R32-2
10
R36-2
9
UK6C
LM324
8
AUD+
AUD+
C27
C27-2
C27-2
100K
R35
R35-2
+5V
5
R35-2
6
UK6B
LM324
7
AUD-
R35-2
100K
R37
AUD-
EDB7
2K
R41
+5V
+5V
UH4
16
VCC
3
4
6
11
13
14
EDB3
EDB2
EDB1
EDB0
D0
D1
D2
D3
D4
D5
2
5
7
10
12
15
Q0
Q1
Q2
Q3
Q4
Q5
SAEN2,3
FLIP
H4-12
H4-15
9
OL
Cp
1
RESET
Mr
GND
74LS174
8
GND
H4-12
150
R30
START_LED1
H4-15
150
R31
START_LED2
STREN
13
12
F4-12
74LS14
F4-12
9
STOP
10
UF5C
74S03
8
EDB7
RESET
1K
R7
+5V
Sheet: /Audio output, output latch and Math Box status/
File: audio.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 10/12



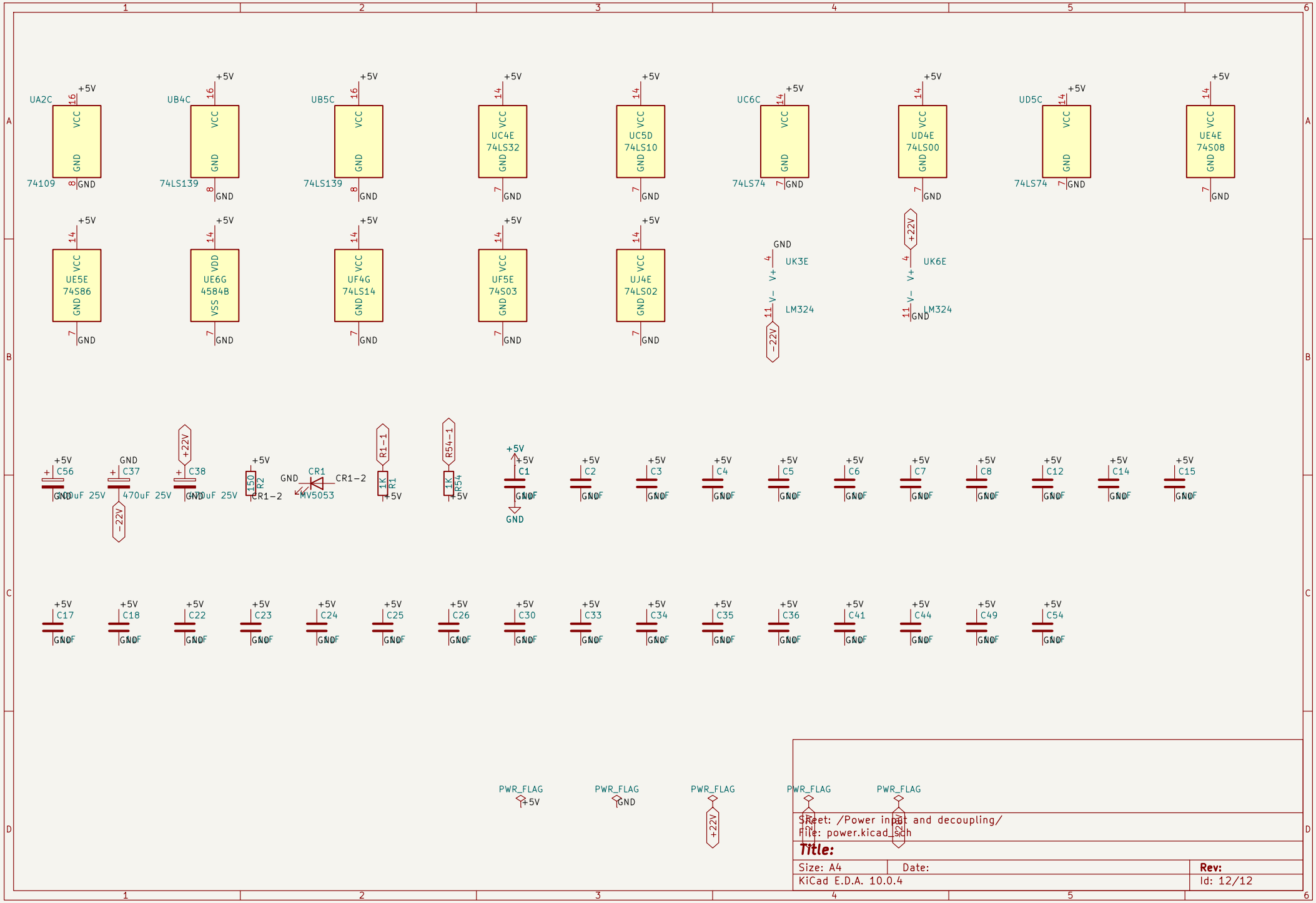




Sheet: /Interconnect, edge connector and test points/
File: io.kicad_sch

Title:

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UA2C
74109
UB4C
74LS139
UB5C
74LS139
UC4E
74LS32
UC5D
74LS10
UC6C
74LS74
UD4E
74LS00
UD5C
74LS74
UE4E
74S08
UE5E
74S86
UE6G
4584B
UF4G
74LS14
UF5E
74S03
UJ4E
74LS02
UK3E
UK6E
LM324
+5V
GND
+22V
-22V
C56
C37
C38
470uF 25V
CR1
CR1-2
MV5053
R2
150
R1
1K
R1-1
R54
R54-1
C1
C2
C3
C4
C5
C6
C7
C8
C12
C14
C15
C17
C18
C22
C23
C24
C25
C26
C30
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C54
PWR_FLAG
Sheet: /Power input and decoupling/
File: power.kicad_sch
Title:
Size: A4
Date:
Rev:
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